Physics c191 Final Project Report

A Brief Introduction to Holographic Error Correction

Keshav B. Deoskar, Jon D. Goker, Matthew J. Thomas

Abstract

Holographic Error Correction is a beautiful idea providing toy models for, along with applications of, the AdS/CFT correspondence in the realm of Quantum Error Correction. In this report, we hope to elucidate the basic ideas of Holography, what makes a code "Holographic", and a concrete example; namely, one of the HaPPY Codes. We attempt to give the big ideas and intuition, but kindly request the reader to consult the references for details in order to satisfy the ten-page requirement. This report assumes familiarity with content covered in the course.

Contribution Statement

M.J.T. contributed to Appendices; J.D.G. contributed heavily to Section 4, 5; K.B.D. contributed heavily to Sections 1, 2, 3, 4.

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1 Introduction

AdS/CFT

In the 1990's, a revolution in string-theory occurred [11] with the discovery of the AdS/CFT correspondence. The idea is that a gravitational theory in $d-$dimensional Anti de-Sitter (AdS) Space is *dual* (i.e. there is a mapping between operators/quantities in the two theories) to a $(d-1)$ Conformal Field Theory (CFT) defined on the boundary of the AdS space. Thus, there is a **holographic correspondence** (as in hologram, being one dimension smaller), between the bulk AdS and the boundary CFT theories.

Quantum Error Correction

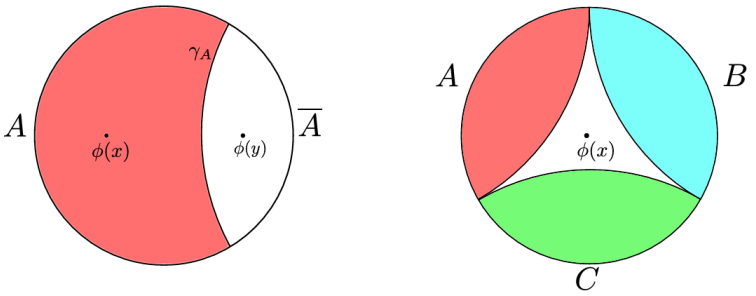
Around the same time, breakthroughs like Shor's code catapulted Quantum Computation into a position of intense interest. It was quickly realized that, like classical computation, error correction is required to make quantum computation practical. Quantum Error Correction Codes (QECCs) give us a way to encode some information i.e. state of, say, k "logical" qubits using, say, n physical qubits such that they are protected against some number d of qubits being malformed/distorted without the logical information being lost, thus forming an $[[n, k, d]]$ code.

Holographic Codes

The AdS/CFT correspondence provides a "dictionary" relating fields in the bulk gravitational theory to operators in the boundary CFT. For a given boundary subregion A, there is an associated **Causal Wedge, $\mathcal{C}[A]$** of points in the bulk defined as the intersection of the bulk-future and bulk-past of the boundary domain of dependence of A. Think of light-cones in Special Relativity; $\mathcal{C}[A]$ is essentially the set of bulk-points that can interact causally with A.

Via the AdS/CFT dictionary, a field value $\phi(x)$ for point x in the causal wedge of boundary-subregion A can be exressed using operators of the CFT which are supported only on A (i.e. zero everywhere else on the boundary), thus letting us "reconstruct" $\phi(x)$ on the boundary. This property where is called **Subregion Duality** as a subregion captures all the information required to reconstruct the bulk-point.

In [4], this reconstruction was shown to extend to a larger set of points called the **Entanglement Wedge, $\mathcal{E}[A]$**.



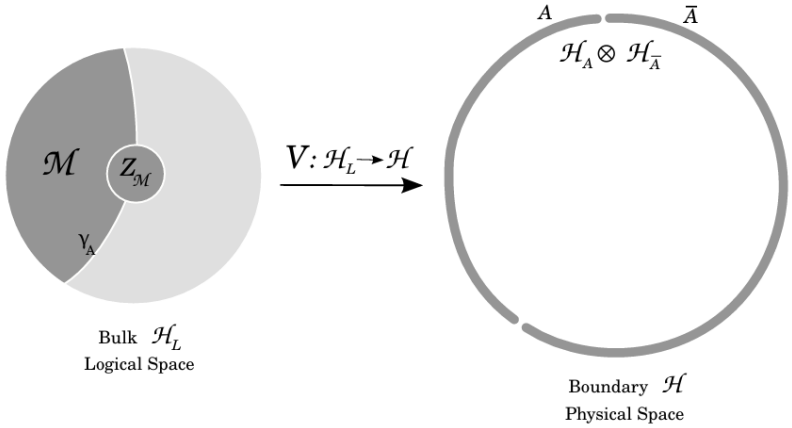
Source: Figure 1, [6].

In the left figure, the shaded region is the intersection of $\mathcal{E}[A]$ with a fixed-time bulk slice. Thus, $\phi(x)$ has a representation in the CFT defined on A and similarly for $\phi(y)$ on $\overline{A}$.

On the right, $\phi(x)$ doesn't lie in the entanglement wedges of A, B, C so it doesn't have a representation on those boundary subregions. However it does lie in the entanglement wedge of AB, BC, AC and so it does have a representation on those subregion CFTs.

The connection to error correction becomes clear when we realize a given point in the bulk can lie in multiple causal wedges! So, there is a "redundancy" very reminiscent of QECCs with the bulk "logical" quantitites being encoded in the boundary "physical" operators with some redundancy giving us protection against errors. Notably, some bulk value $\phi(x)$ can be reconstructed on different subregions of the boundary i.e. it has representations in the CFTs defined on multiple boundary subregions.

Holographic Codes are, in particular, **Erasure Correction Codes** where "erasure" is defined as losing access to a subset of the physical degrees of freedom. We will not discuss other types of errors in this report (eg. bit flips, phase flips, etc.), however they can be handled as well.



Source: Figure 1, [9]

2 What makes a given QECC a "good" Holographic Code?

2.1 Causal and Entanglement Wedges

As discussed earlier, in the AdS/CFT Correspondence, we can reconstruct the field-values at points lying in the Causal or Entanglement Wedges. So, in order for our code to be Holographic it should have **analogues of the causal and entanglement wedges**, and some **reconstruction** mapping.

2.2 Ryu-Takayanagi Formulae

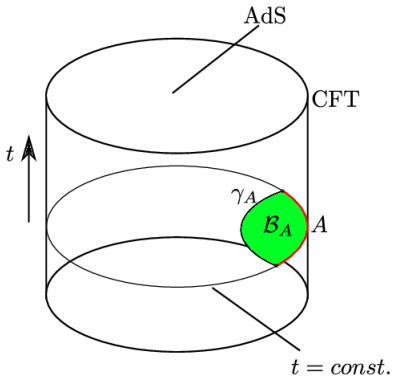
A very important equation in the context of Black Holes, The Bekenstein-Hawking (BH) formula, relates Black hole entropy S_{BH} to the event-horizon (Σ) and Newton's Gravitational Constant G_N as

$$S_{BH} = \frac{\text{Area}(\Sigma)}{4G_N}$$

By the time AdS/CFT was popularized, it had been shown that Black Hole Entropy possessed similar mathematical structure to (Quantum) Entanglement Entropy [5]. So, in 2006, Ryu and Takayanagi conjectured a generalization of the BH Equation in the context of a system with bulk AdS and boundary CFT theories [10]:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N^{d+2}}$$

where A is a space-like manifold on constant time-slice Σ, γ_A is the d-dimensional minimal geodesic whose whose boundary matches the boundary of A, where a geodesic is a line that a body in "free-fall" follows within a given spacetime.



Source: [1]

This equation actually makes some approximations, neglecting a "bulk-entropy" term, but the idea it conveys is that we have

$$\text{entanglement across } A - \overline{A} \ \propto \ \text{Area}(\gamma_A)$$

it and its further refinements are referred to as the **Ryu-Takayanagi (RT) Formula(e)**. RT Formulae give a relationship between Entanglement-Entropy **(quantum mechanical)** and some sort of Area operator **(gravitational/geometrical)**.

Given a CFT state (density matrix) ρ, we can define a boundary state ρ_A on any boundary subregion A. If ρ is "appropriate" then the von Neumann entropy of ρ_A is given by

$$S(\rho_A) = \text{Tr}(\rho \mathcal{L}_A) + S_{bulk}(\rho_{\mathcal{E}_A})$$

where the second RHS term is the bulk-entropy in the Entanglement Wedge $\mathcal{E}[A]$.

It turns out that *every* conventional Quantum Erasure Correction Code has an RT-*like* formula but he area operator does not always reflect interesting "bulk-geometry". We explore this more in the next section.

So, in order for an Error Correcting Code to be "Holographic" we would like it to have **Subregion Duality**, well-defined analogues of **causal/entanglement wedges** and reconstruction, a **non-trivial Area Operator**, ather properties observed in AdS/CFT like Radial Commutativity

3 More on RT Formulae

To get a better feel for RT-Formulae, this section walks through the main ideas in [6].

3.1 An example: The 3-Qutrit Code

This important QECC, encoding one "logical" qutrit into a hilbert space of three "physical" qutrits was initially introduced as a model of Holography. The code-subspace $\mathcal{H}_{code}$ is spanned by the basis

$$|\widetilde{0}\rangle \equiv \frac{1}{\sqrt{3}}\left(|000\rangle + |111\rangle + |222\rangle\right), \ |\widetilde{1}\rangle \equiv \frac{1}{\sqrt{3}}\left(|000\rangle + |111\rangle + |222\rangle\right), \ |\widetilde{2}\rangle \equiv \frac{1}{\sqrt{3}}\left(|021\rangle + |102\rangle + |210\rangle\right)$$

It turns out (see [3] for details) there exists a unitary U_{12} on the physical hilbert space *supported only on the first two physical qutrits* such that we have

$$U_{12}^{\dagger}|\widetilde{i}\rangle_1|\chi\rangle_{23}, \text{ for } |\chi\rangle \equiv \frac{1}{\sqrt{3}}\left(|00\rangle + |11\rangle + |22\rangle\right)$$

The fact that such a unitary U_{12} exists allows us to protect the state of the logical qutrit against the erasure of the third physical qutrit. Why?

If the single-qutrit state we *want* to protect is $|\psi\rangle = \sum_{i=0}^{2} C_i|i\rangle$ and we encode it in a single-qutrit, then the state is easily corruptable. But if we use the 3-qutrit encoding to define a logical state $|\widetilde{\psi}\rangle = \sum_{i=0}^{2} C_i|\widetilde{i}\rangle$ then even if the third physical qutrit used in the encoding is lost, we can apply $U_{12}^{\dagger}$ to the first two physical qutrits to recover the state $|\psi\rangle$ on the first qutrit.

$$U_{12}^{\dagger}|\widetilde{\psi}\rangle = |\psi\rangle_1|\chi\rangle_{23}$$

The symmetry in the definition of the basis states for $\mathcal{H}_{code}$ ensures that similar unitaries U_{23} and U_{13} exist as well, which means that $|\psi\rangle$ can be recovered **as long as any two of the qutrits remain un-erased**.

RT Formula for 3-Qutrit Code

Suppose we have an arbitrary mixed state $\widetilde{\rho}$ on $\mathcal{H}_{code}$ encoding the "logical" mixed state ρ. Then, we can write

$$\widetilde{\rho} = U_{12}\left(\rho_1 \otimes (|\chi\rangle\langle\chi|)_{23}\right)U_{12}^{\dagger}$$

So defining $\widetilde{\rho}_3 \equiv \mathrm{Tr}_{12}\widetilde{\rho}$ and $\widetilde{\rho}_{12} \equiv \mathrm{Tr}_3\widetilde{\rho}$ we have the von-Neumann Entropies

$$S(\widetilde{\rho}_3) = \log(3), \quad S(\widetilde{\rho}_{12}) = \log(3) + S(\widetilde{\rho})$$

- If we define area operators $\mathcal{L}_3 = \mathcal{L}_{12} \equiv \log(3)$ then the RT Formula seems to hold for this code.
- We have **subregion duality** in the sense that state $|\phi\rangle$ can be recovered using only a subset of the physical qutrits.

However, notice that the "Area Operator" $\mathcal{L}$ is a constant. Intuitively, this would correspond to our "geodesic" being a straight line and our bulk-geometry being flat/non-interesting.

3.2 Conventional Quantum Erasure Correction

In section 3 of [6], Theorem 3.1 gives conditions to predict whether or not a code can correct a given erasure. Roughly, it's shown that for a system with a factorizable (finite dim.) hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\overline{A}}$ and $\mathcal{H}_{code} \subseteq \mathcal{H}$, any logical operator on $\mathcal{H}_A$ can be represented by an operator on A if and only if we can recover the full state of the code subspace on some A_1 by applying an operator $U_A^\dagger$ (where $\mathcal{H}_A = (\mathcal{H}_{A_1} \otimes \mathcal{H}_{A_2}) \oplus \mathcal{H}_{A_3}$ and appropriate conditions on $|A_1|, |A_2|, |A_3|$).

In the 3-Qutrit code, for example, $\mathcal{H}_{12}$ and $U_{12}^\dagger$ would be analogous to $\mathcal{H}_{A_1}$ and $U_A^\dagger$. There is a $|\chi\rangle$ state analogous to the $|\chi\rangle$ defined in the 3-Qutrit case which allows us to carry out the reconstruction due to the way it entangles the basis states.

Importantly, it's shown in section 3.2 [6] that there exist RT-like formulae

$$S(\widetilde{\rho}_A) = S(\chi_{A_2}) + S(\widetilde{\rho})$$
$$S(\widetilde{\rho}_{\overline{A}}) = S(\chi_{A_2})$$

which are reminiscent of the RT-Formula but with with $\mathcal{L}_A = S(\chi_{A_2}) I_{code}$. Although it's not perfect, the area term does still seem to arise due to the entanglement in the $|\chi\rangle$ state.

All is not well just yet!

An issue however, is that the "bulk-entropy" term does not appear symmetrically in $S(\widetilde{\rho}_A)$ and $S(\widetilde{\rho}_{\overline{A}})$. It turns out this problem arises because we demand that the entire state (and all of its observables) be recoverable on A.

3.3 Subsystem Quantum Erasure Correction

A generalization of convention Quantum Erasure correction is when we consider our code-space to factorize as $\mathcal{H}_{code} = \mathcal{H}_a \otimes \mathcal{H}_{\overline{a}}$ and then **only ask for recovery of states in $\mathcal{H}_a$**.

In AdS/CFT, we consider systems possesssing **complementary recovery** i.e. if we can recover arbitrary $\widetilde{O}_a$ on A we can also recover arbitrary $\widetilde{O}_{\widetilde{a}}$ on $\overline{A}$. The motivation behind this assumption is related to "Quantum Mutual Independence"; see footnote 4 of Section 4.1 [6].

Section 4.2 of [6] shows that considering a *subsystem erasure code* with *complementary recovery* fixes the asymmetry in the RT formulae!

However there is still a problem!

Even in this case, the area operator is trivial i.e. proportional to the identity (see [6])

$$\mathcal{L}_A \otimes I_{\overline{a}} = I_a \otimes \mathcal{L}_{\overline{A}} = S(\chi_{A_2}) I_{code}$$

The reason for this is that we assume our codespace factorizes as $\mathcal{H}_{code} = \mathcal{H}_a \otimes \mathcal{H}_{\overline{a}}$ and the only operators of $\mathcal{H}$ that can be shared between $\mathcal{H}_a, \mathcal{H}_{\overline{a}}$ are multiples of the identity. This, combined with Complementary Recovery, is a very strict condition and constrains the area operator to be trivial.

To fix this, we must generalize to a case where the bulk algebras of operators on A, $\overline{A}$ can have more in common. So, we drop the assumption that every bulk-observable is recoverable and instead consider just a **subalgebra** which constitutes a **von Neumann Algebra**.

3.4 Operator-Algebra Quantum Erasure Correction

Section 5 of [6] shows that if we work with a

1. Subsystem Erasure Code... with
2. Complementary Recovery... and
3. We only demand that a subalgebra $M \subseteq \mathcal{H}_{code}$ is reconstructable on the boundary

then indeed we have non-trivial area operators and the RT-Formula holds for a region A if and only if we can reconstruct any operator in M on A.

We only consider M being a von Neumann algebra because any set of **observables** forms a von Neumann algebra (since such a set must be closed under addition, multiplication, scalar multiplication, and complex conjugation).

Here, M' is the commutant i.e. the set of linear operators on $\mathcal{H}$ which commute with every $O \in M$. The set $Z_M \equiv M \cap M'$ is called the **center** of M.

The definition of **complementary recovery** in this case is that if we can represent any operator in M on A then we can represent any operator in M' on $\overline{A}$.

The intuition behind this is that bulk-operators representable on $\overline{A}$ can be written only using operators supported on $\overline{A}$ (zero everywhere else) so they should commute with operators supported on A, and thus with bulk-operators representable on A.

Erasure correction with the above assumptions is known as **Operator-Algebra Quantum Erasure Correction**. More formally, the result is

Theorem 1.1. from [6]

Say we have a finite-dimensional Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\overline{A}}$, a code subspace $\mathcal{H}_{code} \subseteq \mathcal{H}$ and a von-neumann Algebra M (and its commutant is denoted M') acting on $\mathcal{H}_{code}$. Then, the following statements are equivalent:

- There exists an operator $\mathcal{L}_A \in Z_M \equiv M \cap M'$ such that, for any state $\widetilde{\rho}$ on $\mathcal{H}_{code}$, we have

$$S(\widetilde{\rho}_A) = \mathrm{Tr}(\widetilde{\rho}\mathcal{L}_A) + S(\widetilde{\rho}, M)$$
$$S(\widetilde{\rho}_{\overline{A}}) = \mathrm{Tr}(\widetilde{\rho}\mathcal{L}_A) + S(\widetilde{\rho}, M')$$

- For any operators $\widetilde{O} \in M$, $\widetilde{O}' \in M'$, there exists operators $O_A, O'_{\overline{A}}$ on $\mathcal{H}_A, \mathcal{H}_{\overline{A}}$ respectively

such that, for any state $|\widetilde{\psi}\rangle \in \mathcal{H}_{code}$, we have

$$O_A|\widetilde{\psi}\rangle = \widetilde{O}|\widetilde{\psi}\rangle \; O_A^\dagger|\widetilde{\psi}\rangle = \widetilde{O}^\dagger|\widetilde{\psi}\rangle$$
$$O_{\overline{A}}|\widetilde{\psi}\rangle = \widetilde{O}|\widetilde{\psi}\rangle \; O_{\overline{A}}^\dagger|\widetilde{\psi}\rangle = \widetilde{O}^\dagger|\widetilde{\psi}\rangle$$

Having M be a von Neumann Algebra on $\mathcal{H}$ allows us to use the *Wedderburn Decomposition* [9, See Chapter 3] for $\mathcal{H}, M, M', Z_M$

$$\mathcal{H} = \left[\oplus(\mathcal{H}_{A_\alpha} \otimes \mathcal{H}_{\overline{A}_\alpha})\right] \oplus \mathcal{H}_0, \; M = \left[\oplus(\mathcal{L}(\mathcal{H}_{A_\alpha}) \otimes \mathcal{H}_{\overline{A}_\alpha})\right] \oplus \mathbf{0}$$
$$M' = \left[\oplus(\mathcal{H}_{A_\alpha} \otimes \mathcal{L}(\mathcal{H}_{\overline{A}_\alpha}))\right] \oplus \mathbf{0}, \; Z_M = \oplus_\alpha \left(c_\alpha I_{A_\alpha} \otimes I_{\overline{A}_\alpha}\right)$$

where $\mathcal{H}_0$ is the null space, and $\mathbf{0}$ is the zero operator on $\mathcal{H}$, and $\mathcal{L}(\cdot)$ denotes the set of linear operators on the set $\cdot$).

Using this, [6] shows that there are RT-Formulae

$$S(\widetilde{\rho}_A) = \mathrm{Tr}(\widetilde{\rho}\mathcal{L}_A) + S(\widetilde{\rho}, M)$$
$$S(\widetilde{\rho}_{\overline{A}}) = \mathrm{Tr}(\widetilde{\rho}\mathcal{L}_{\overline{A}}) + S(\widetilde{\rho}, M')$$

with **non-trivial area operator**

$$\mathcal{L}_A = \oplus_\alpha S(\chi_{A_2^\alpha}) I_{a_\alpha \overline{a}_\alpha}$$

and $S(\widetilde{\rho}, M)$ being a generalization of the bulk-entropy called the **algebraic entropy** defined in the langauge of von Neumann algebras.

4 HaPPY Codes

Now, we discuss a more concrete example of holographic codes: The HaPPY codes. These QECCs provide exactly solvable toy models that provide a discretized analog to a vacuum AdS/CFT correspondence. We will roughly follow the exposition in [8].

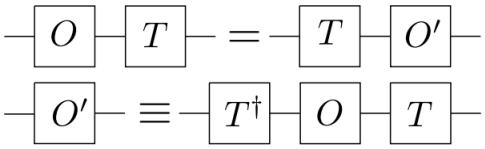
4.1 Perfect Tensors and Tensor Networks

HaPPY Codes are built from **perfect tensors** contracted together to form networks that tile a hyperbolic plane (2D surface of negative curvature, like AdS-Space).

A 2n-index tensor $T_{a_1 a_2 \ldots a_2 n}$ is **Perfect** if, for any bipartition of its indices into a set A and complementary set A^c with $|A| \leq |A^c|$, T is proportional to an isometric tensor from A to A^c.

where a tensor $T : \mathcal{H}_A \to \mathcal{H}_B$ is **isometric** if it preserves the inner product, or equivalently, if $TT^\dagger = I_{\mathcal{H}_A}$ and $T^\dagger T$ projects $\mathcal{H}_B$ to the range of T.

Basically, for any bipartition of the indices, we can reconstruct the rest of the indices (or alternatively any bipartition of spins is maximally entangles with the remaining spins). A simple explicit construction of this is the 5 qubit code, a six index perfect tensor. This is more explicitly explored in the appendix. For now it suffices to keep this image in ones head. The most significant property of such isomorphisms is the ability to push operators through such isometries as shown;

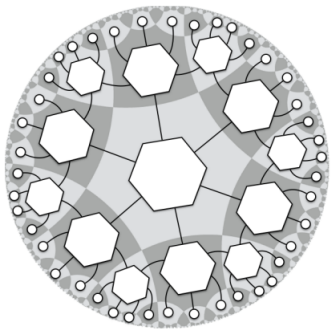


Source: Figure 1, [8]

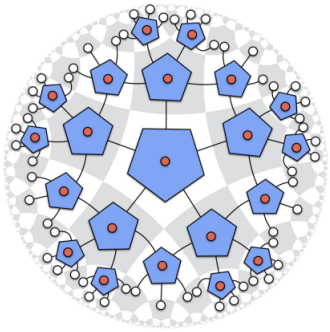
Here we see that for an operator acting on some index (take it as logical for the context of these codes) we can reconstruct the operator on other legs of the tensor (take these as physical spins). There exists a redundant encoding scheme for an operator acting on any index in any tensor in a network of tensors. This is the crux of holographic error correction and the central property we make use of for HaPPY codes.

4.1.1 Construction of Holographic States and Codes

Using perfect tensors, we will construct our *Holographic States/Codes*. To do so, we want to consider a network of tensors contracted with eachother such that the state of the network is either encoded in or determined by the boundary spins. Specifically, we will focus on different tilings of 2 dimensional hyperbolic geometries as the analogs of Cauchy slices. These tilings are chosen as they provide a discrete scale invariant construction that allows for any boundary operator to be reconstructed in the center while preserving the local structure of the tiling. Consider the following two tilings of a holographic state and code.



(a) Holographic hexagon state



(b) Holographic pentagon code

Each polyon in the figures above represents one perfect tensor, and a line between two polygons represents one index from each tensor being contracted. The dots represent *uncontracted* indices (red circles in the bulk, representing logical d.o.f.; white circles on the boundary, representing physical d.o.f).

Since they are perfect tensors, the polygons represent isometries, and since we recognize a product of isometries is an isometry, we view this as an isometric encoding of the bulk spins to those on the boundary. We recognize as we increase the number of isometries in the bulk, the rate of the code asympotically approaches ≈ 0.447 logical to physical spins.

4.2 Entanglement Structure for Holographic States

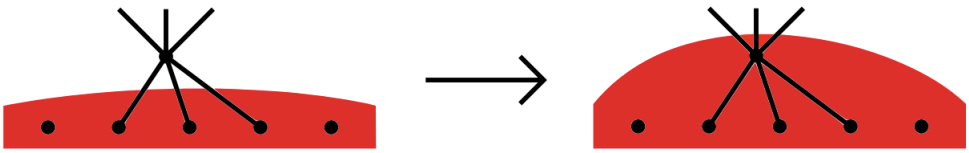
While we have thus far constructed a code that seems to display holographic properties, we want to explicitly confirm if they display more of the qualities of holography so they can be considered a valid toy model. In this section we will explore some such methods.

4.2.1 RT Formula For Tensor Networks

Previously we have outlined the Ryu-Takayanagi formula as a hallmark of holography in terms of quantum information. We now want to find a way to construct γ_A in a way that defines it properly by the above prescription. Let γ_A be a cut through our tensor network, dividing the space into two sets of disjoint tensors. However since the product of isometries is itself an isometry, we can represent each disjoint region as a tensor, P and Q respectively, with a contraction along the cut. For a set of boundary legs and its complement, A and A^c respectively, we define the boundary of the cut as matching the boundary of A if uncontracted legs of P are those that define A. As such we can define the minimal bulk geodesic γ_A as the cut of shortest distance with matching boundaries as defined in the last sentence.

To actually find the minimal geodesic γ_A corresponding to $A = \bigcup_i A_i$ for A_i being connected boundary subregions, we use the **greedy algorithm**:

- Beginning at the boundary, A_i, we consider a sequence of cuts $\{c_\alpha\}$ and accompanying tensors $\{P_\alpha\}$ where each index is related to the previous one by a step on the lattice or an iteration of the code.
- In each step we add to our P_α any isometry with at least half of its legs contracted with the spins on the current geodesic. We repeat this until it cannot be continued locally (no more such isometries).
- Using each one of the γ_{A_i}'s we define the γ_A^* as the union of all these gammas and use them for our RT formula of disjoint regions.



One iteration of the Greedy Algorithm via [8]

Even when the prerequisites for this formula are not fullfilled (simply-connected planar tensor network of perfect tensors, whose graph has "non-positive curvature"), this will still provide a lower bound on the entropy.

Additionally, we can construct the tripartite information and prove (done in [8]) for any partition of a perfect tensor states' $2n$ indices into 4 disjoint sets of non-negative cardinality, we have

$$I_3(A, B, C) < 0$$

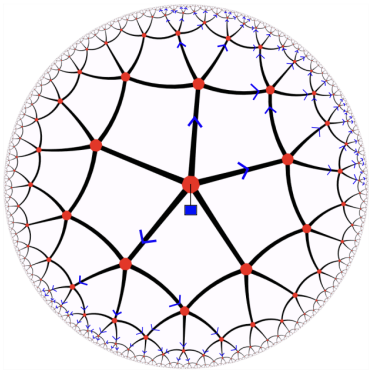
This provides another confirmation that our code mimics the behavior of holography as expected, as [7] showed that in the full holographic setting, $I_3 < 0$ follows from the RT formula.

4.3 Quantum Error Correction in Holographic Codes

Now, given the confirmed similar behavior between our codes and holography, we want to try and see the error protection properties of AdS CFT manifest in our codes. Namely, it would not make sense for an erasure error to exist in our conformal theory, so we'd like for an analogous case to exist for our holographic codes in the discrete case.

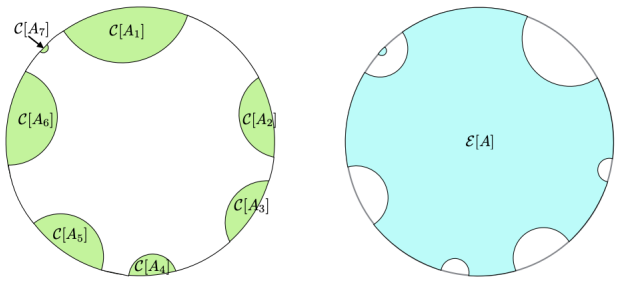
4.3.1 Bulk Reconstruction From Tensor Pushing

Since we have foliated our space with isometric perfect tensors all of which are equipped with a notion of tensor pushing it is clear that we have an ability to reconstruct from a boundary region logical spins deep inside the bulk. This is shown explicIty below in the first figure where the blue square in the center is pushed to the boundary (which can be done in reverse) shown by blue arrows. To make this more concrete, we define a **Causal Wedge** as the bulk region that is enclosed by applying the greedy algorithm. However, when we take into account disjoint boundary regions, we recognize that greedy geodesic could potentially go too far for the definition of a causal wedge, and thus must define an analogy of an entanglement wedge as follows. The **Greedy Entanglement Wedge** is the set of points reached by applying the greedy algorithm to all connected components of boundary A. This allows us to define a subset of the bulk that allows for reconstruction in an analysis construction to classical AdS as we had originally set out to do!



Here we see a general prescription for how reconstruction is performed, pushing from the blue square outward to the boundary. However here the greedy algorithm applied to disconnected regions reaches the center of the bulk motivating the definition of a causal wedge.

Source for figures: [8]



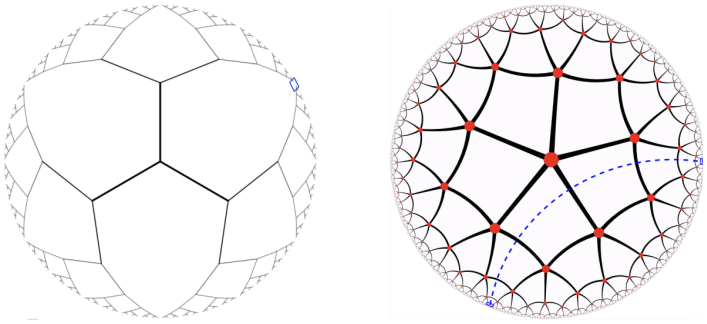
(a) Shallow causal wedge (b) Deep entanglement wedge

Difference between causal wedge and entanglement wedge showing the power of reconstruction when the union of the boundary regions is large, but the disconnected components themselves are small.

4.3.2 Erasure Threshold

We are confident now that we have a working discretized model for the holography in terms of Quantum Tensor Networks, however we now want to actually confirm the error correcting properties. We want to define this in terms of erasure threshold. If there exists some probability p_c that our erasure probability is smaller than, the entanglement wedge will contain the necessary spins for

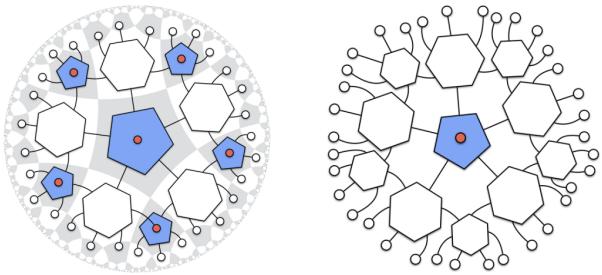
reconstruction with success probability deviating from one that becomes doubly exponentially small as the radius of our space increases. However, this is not the case for all holographic codes as we show in the images below. Namely, if we don't have a code that increases with system size (or alternatively there is a bulk index supported by a constant number of boundary spins) then we cannot have a reasonable error threshold.



(a) Triangle code (b) Pentagon code

For both these examples we see that reconstruction is not possible since we have a central bulk operator defined by 2 or 4 boundary spins respectively. This causes the errors to propogate and make error correction impossible

A modification of the pentagon code, shown below has a valid error correction threshold since we can see as we increase the radius (add mode pentagons) the redundancy increases.



(a) Pentagon/Hexagon code (b) One qubit code

Each of these codes represent holographic codes where we can define a notion or erasure threshold since their redundancy grows with the size of the network

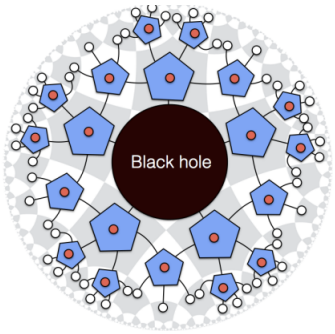
Source for figures: [8] Through calculations included in the appendix B, we will show that for this code we have a lower bound on this error threshold of $p_c \geq \frac{1}{12}$ however numerically it can be shown that $p_c = \frac{1}{2}$ as we would hope and expect for such a maximally entangled system.

5 Future Directions and Discussion

Having now successfully found a discrete toy model that seems to embody the error protecting properties of AdS/CFT we are motivated to explore alternate directions we can take such work into and clarifications to be solidified.

5.1 Black Holes and Holography

In order to construct an analogy to a black hole, we consider the pentagon code, but remove the center bulk index leaving us with 5 uncontracted legs. This increases the size of the bulk subspace of the boundary Hilbert space, which we would expect as we now have black hole states to encode in this model (or more simply put there is now mass). Similarly, as we construct larger and larger black holes, we see that the reconstructable bulk indices decrease, which we would once again expect. A further discussion of black hole interior reconstruction is discussed in [2] and other texts in a more physically motivated manner, however this is out of scope of this discussion.



Representation of a black hole in Vaccum AdS for our tensor network representation
Source: [8]

6 Acknowledgements

We thank the instructors for organizing this insightful course, and particularly appreciate the enlightening discussions with (and patience of) Geoffrey Penington and Thilo Scharnhorst.

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A RT Formula for Tensor Networks

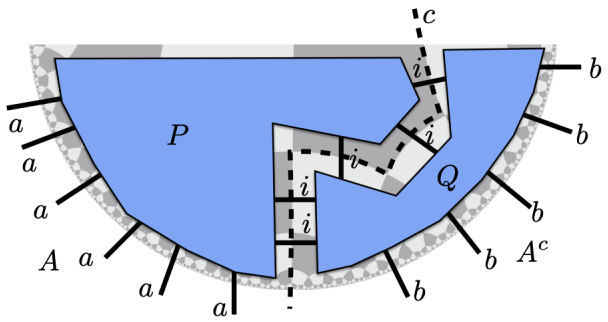
Using the definitions in section 4.2.1, we sketch out the proof that γ_A is indeed the minimal geodesic we are hoping for in the RT formula. Let P and Q be tensors defined by a cut c whose boundary matches that of A. Then we define the holographic state $|\psi\rangle$

$$|\psi\rangle = \sum_{a,b,i} |ab\rangle P_{ai} Q_{ai} \equiv \sum_i |P_i\rangle_A |Q_i\rangle_{A^c}$$

Now to find the density matrix for just one region, we trace out the complementary region as follows

$$\rho_A = \sum_{i,i'} \langle Q_i || Q_{i'} \rangle |P_i\rangle |P_{i'}\rangle$$

By inspection we see that this sum and thus the rank of this matrix will be bounded above by $v^{|c|}$ where v is the number of indices of the tensors in the tiling and c refers to the number of contracted legs across c. See the figure below; As such we see that if $|c| = \gamma_A$ we will have a maximum Von Neumann entropy, as predicted in the RT formula that is proportional to γ_A. One of our checks for this models validity for AdS CFT is indeed correct. (Note: firslty we suppress the $log(v)$ term by considering the entropy in terms of a similar log.Secondly, this proof follows from [8] and is explored deeper in their appendix B). For disjoint boundary regions, this formalism does not hold.



Reference image for proof of RT for Tensor Networks from [8]

B Perfect Tensor Examples

B.1 5-Qubit extension to 6-Qubit

The 5-Qubit code is the smallest error correcting code that protects a single qubit against 2 erasures. The [[5,1,3]] code is a stabilizer code defined by 4 independent stabilizers acting on 5 physical qubits. The stabilizer sub-group is given by $S =< S_1, S_2, S_3, S_4 >$ with source([Preskill]):

$$\begin{aligned} S_1 &= X \otimes Z \otimes Z \otimes X \otimes I, \\ S_2 &= I \otimes X \otimes Z \otimes Z \otimes X, \\ S_3 &= X \otimes I \otimes X \otimes Z \otimes Z, \\ S_4 &= Z \otimes X \otimes I \otimes X \otimes Z. \end{aligned}$$

and $S_1 S_2 S_3 S_4 = Z \otimes Z \otimes X \otimes I \otimes X$. A non trivial choice of logical operators are given by: $\overline{X} = X^{\otimes_5}$ and $\overline{Z} = Z^{\otimes_5}$ which commute with all stabilizers but anti-commute with each other.

Any part of $\overline{X}$ and $\overline{Z}$ must act on at least 3 physical qubits, so losing up to 2 qubits still makes it possible to perfectly restore the logical state. To build a single 6-qubit state that contains the 5-qubit encoding and maximally entangled, a sixth qubit is appended by extending the stabilizers to $S' =< S'_1, S'_2, S'_3, S'_4, S'_5, S'_6 >$:

$$\begin{aligned} S'_1 &= S_1 \otimes I, \\ S'_2 &= S_2 \otimes I, \\ S'_3 &= S_3 \otimes I, \\ S'_4 &= S_4 \otimes I, \\ S'_5 &= \overline{X} \otimes X, \\ S'_6 &= \overline{Z} \otimes Z \end{aligned}$$

where there is now duplicated generators with $S'_j = S_j \otimes I$, $(j = 1, ..., 4)$ and two new constructed stabilizers S'_5 and S'_6. The unique non-zero vector $|\Psi\rangle$ with eigen value $+1$ of the 6 stabilzers is:

$$\begin{aligned} S'_j|\Psi\rangle &= |\Psi\rangle \\ |\Psi\rangle &= |\tilde{0}\rangle \otimes |0\rangle + |\tilde{1}\rangle \otimes |1\rangle \end{aligned}$$

where $|\tilde{0}\rangle, |\tilde{1}\rangle$ are the two logical basis states of the 5 qubit code ($S_j|\tilde{b}\rangle = +1|\tilde{b}\rangle$, $b = 0, 1$. Viewing $|\Psi\rangle$ as a rank 6 tensor, any subset of three or fewer indices has a reduced density matrix proportional to the identity ($\rho_A \propto I_A$ with $|A| \leq 3$) so it is a perfect tensor. This perfect tensor condition means erasure up to n-1 legs can always be recovered (here n = 3). Tiling hyperbolic space (like seen in AdS/CFT) with such perfect tensors yields a tensor-network isometry that shows a holographic code.

C Error Threshold

In certain holographic codes, there exists a critical erasure rate ϵ^* such that if boundary qubits are randomly lost with a probability less than ϵ^*, the central bulk qubit can almost certainly be reconstructed. The analysis for holographic codes is difficult because of the geometry but a hierarchical analysis can still be done to understand the failure probability.

C.1 Analytic Upper Bound with Single Qubit Hexagon Code

This code starts with a single [[5,1,3]] tensor and builds up n [[6,0,4]] tensors from it. So there is a central pentagon surrounded by n layers of hexagons and two worst-case error scenarios at level j, where s_j si the chance a single tensore at layer j fails to be recovered, and d_j is the chance two adjacent tensors both fail. These obey:

$$\begin{aligned} \mathrm{s}_(j+1) &\leq 3d_j + 3s_j^2 \\ \mathrm{d}_(j+1) &\leq 3_{sj}(3d_j + 3s_j^2) \end{aligned}$$

Solving these inequalities shows a threshold:

$$\epsilon^* = 1/12, \lambda = \frac{1+\sqrt{5}}{2}$$

such that when $\epsilon \leq \epsilon^*$, the probability p of successfully recovering the center is:

$$p \geq 1 - \varepsilon_* \left(\frac{\varepsilon}{\varepsilon_*} \right)^{\lambda n}$$

C.2 Numerical Bound

From simulating many instances of random erasure on different tilings, a clear threshold is found $p \leq p^c_{\text{greedy}} \approx 0.26$, and $p^c_{\text{greedy}} \approx 0.5$